DE–2800

Sub. Code
101

DISTANCE EDUCATION

M.C.A.(S) DEGREE EXAMINATION, DECEMBER 2015.

First Semester

DIGITAL COMPUTER ORGANISATION

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Convert the following.

   (a) $(Af20.10D)_{16}$ into its binary equivalent.

   (b) $(175)_8$ into its decimal equivalent.

2. State various boolean laws.

3. Explain ring counter with diagram.

4. Explain the following in detail.

   (a) Static RAM

   (b) Dynamic RAM.

5. Draw the circuit of a shift register and explain its working principles in detail.

6. How a synchronous counter differ from asynchronous counter? Explain.

7. What is interface? Why interfacing devices are needed? Explain.

8. What are interrupt? Why are they used? Explain what happens when a computer recognizes an interrupt?

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Obtain the simplified expression in SUM of PRODUCT form

(a) $f(A,B,C)=\sum(3,4,5,6,7)$

(b) $f(A,B,C,D)=\sum(4,6,7,15)$

10. Explain Binary adder –subtractor with a neat diagram.

11. Simplify the expression using k-map.

$$f(A,B,C,D,E)=\sum(0,1,\ 2,\ 4,\ 7,\ 8,\ 12,\ 14,\ 15,\ 16,17,18,\ 20,\ 24,\ 28,\ 30,31)$$

12. Draw the schematic diagram of a master-slave J–K flipflop. Discuss its working principle. What are its advantages over type of f.f?

13. What is a counter? Design a mod-5 counter with any one of the flip-flop type R-S/J-K.

14. Explain in detail the instruction cycle.

15. (a) Explain in detail about maskable interrupt.

(b) Explain DNA in detail.

DE–2801

Sub. Code
102

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, DECEMBER 2015.

First Semester

C AND DATA STRUCTURES

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Compare the following pairs of statements with respect to their syntax and function.

   (a) break and continue

   (b) goto and break

2. Differentiate the following :

   (a) 'call by value' and 'call by reference'.

   (b) 'do...while loops' and 'while loops'.

3. What is a pointer? How is it declared? Write a C program to reverse a string using pointers.

4. What is dynamic memory allocation? Explain the various memory allocation functions with its task.

5. Design an algorithm to generate n^{th} number of fibanacci sequence.

6. Explain the enqueue and dequeue operations using array.

7. Write an algorithm to insert a node into a doubly linked list.

8. Discuss the two types of binary tree representation.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain the various decision making statements in 'C' Language with example.

10. Differentiate pointer from array? Write a C program to display the contents of an array using a pointer arithmetic.

11. Explain the various file operations in C language with example.

12. Write a program to check whether an applicant is eligible to admit in MCA programme offered by a college according to the following criteria.

    (a) Maths marks > = 80

    (b) Physics marks > = 75

    (c) Chemistry marks > = 70

    (d) Aggregate marks in all three subjects > = 80

13. Write an algorithm to convert infix expression to postfix expression. Implement the algorithm using one example and show the stack Contents.

14. Explain the linked list implementation of stack ADT in detail.

15. (a) Write an algorithm to determine the number of elements in a tree.

    (b) Write an algorithm to search a particular data in a singly linked list.

DE–2802

Sub. Code
103

DISTANCE EDUCATION

M.C.A. (S) DEGREE EXAMINATION, DECEMBER 2015.

First Semester

RELATIONAL DATABASE MANAGEMENT SYSTEMS

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE of the following questions.

1. What is meant by schema and subschema? Explain.

2. Explain the mapping operation in relational model.

3. Discuss the properties of relational database.

4. List out the various types of SQL commands for data access with example.

5. How will you modify the structure of a table? Discuss with an example.

6. Explain the features of embedded SQL statements with an example.

7. List the benefits of oracle.

8. What do you meant by query? Explain the concept of sub queries with example.

PART B — (4 × 15 = 60 marks)

Answer any FOUR of the following questions.

9. What is meant by DBMS? Explain its features and advantages.

10. Discuss the concept of data control facilities in detail.

11. Illustrate the overview of oracle database architecture with neat diagram.

12. Describe the concept of expressions in PL/SQL with suitable examples.

13. Explain the various privileges commands with various options.

14. Define constraint. Describe the various types of constraints available in oracle with example.

15. Define data modeling. Explain various types of data models.

DE–2803

Sub. Code
104

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, DECEMBER 2015.

First Semester

OPERATING SYSTEMS

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Describe the important services of an operating system.

2. Distinguish between the preemptive and non-preemptive scheduling. Explain each type with an example.

3. What is deadlock? Explain the necessary conditions for a deadlock to occur.

4. Explain the LRU page replacement algorithm with an example.

5. Illustrate the various file access methods in detail.

6. Discuss the essential properties of the following types of operating system.

   (a) Time sharing

   (b) Real time

   (c) Clustered.

7. What is meant by paging? Discuss.

8. How file system performance can be measured? Explain with an example.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain the types of system calls provided by a typical operating system.

10. Define process. Explain the process states and process control block with a neat sketch.

11. Consider the following set of processes, with the length of the CPU burst given in milli seconds :

Process	Burst Time	Priority
P1	10	3
P2	1	1
P3	2	3
P4	1	4
P5	5	2

The processes are assumed to have arrived in the order P1, P2, P3, P4, and P5, all at time 0.

(a) Draw four Gantt charts that illustrate the execution of these processes using the following scheduling algorithms : FCFS, SJF, non preemptive and priority algorithm.

(b) What is the turnaround time of each process for each of the scheduling algorithms in part (a)?

(c) What is the waiting time of each process for each of the scheduling algorithms in part (a)?

(d) Which of the algorithms in part (a) results in the minimum average waiting time (overall processes)?

12. Explain any two deadlock avoidance algorithms with an illustration.

13. Describe the FIFO, LRU and second chance page replacement algorithms with an example.

14. Discuss the various file allocation methods with suitable example.

15. What are the goals of I/O software? Explain in detail.

DE–2804

Sub. Code
201

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, DECEMBER 2015.

Second Semester

OBJECT ORIENTED PROGRAMMING AND C++

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Briefly explain the characteristics of OOPS.
2. Write the structure of C++ program. Give an example.
3. State the use of scope resolution operator? Illustrate with an example.
4. Differentiate 'call by value' from 'call by reference'.
5. Discuss the use of virtual function with example.
6. State the uses of copy constructor.
7. Explain the hierarchical inheritance with an example.
8. Write a procedure to delete the end of a file.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain the data types available in C++.
10. Write a C++ program to multiple any two matrices with order N×M.

11. What is an inline function? State the advantages and disadvantages of inline function.

12. (a) Describe the importance of destructors. (7)

    (b) Briefly explain the type conversion. (8)

13. Explain the virtual base class and member classes.

14. Write a short note on :

    (a) 'This' pointer (8)

    (b) Standard manipulator. (7)

15. What are the functions used to handle file pointers? Explain them with examples.

DE–2805

Sub. Code
202

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, DECEMBER 2015.

Second Semester

UNIX AND SHELL PROGRAMMING

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Explain the devices and directories.

2. Explain the grip family files in detail.

3. What is meant by shell? How to execute shell scripts?

4. Explain looping statement through illustration.

5. Describe I/o redirection commands with suitable example.

6. What is the purpose of 'cal' and 'zap' commands?

7. Explain the error recovery.

8. Explain the features of tbl and eqn pre processes.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. List the features of UNIX operating system.

10. Explain advanced filter commands and its application.

11. Explain the use of meta characters and command line arguments character with example.

12. Write a shell program to display the following fibonaci series. 1,1,2,3,5,8,13,21.

13. Describe in detail about the Unix system calls.

14. Explain the use if functions and procedures in shell programming with suitable examples.

15. Write short note on the following :

    (a) Different shells (5)

    (b) Sed (5)

    (c) Filters. (5)

DE–2806

Sub. Code
203

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, DECEMBER 2015.

Second Semester

COMMUNICATION SKILLS

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. How could you improve your art of writing in English?

2. What are communication aids?

3. Bring out the importance of listening skill.

4. Write a note on body language.

5. Comment on mack group discussion.

6. How are communication skills important for meetings?

7. What is semantics?

8. Write a note on model presentation.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Discuss interpersonal communication.

10. Trace the commandments of communication.

11. What are the different types of interview?

12. Enumerate the process and purpose of group discussion.

13. Describe negotiation techniques.

14. Distinguish upward and downward communication.

15. Illustrate planning and designing.

DE–2807

Sub. Code
204

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, DECEMBER 2015.

Second Semester

COMPUTER GRAPHICS

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. What do you mean by computer graphics? List the applications.

2. Describe the working principle of light pen device.

3. What is transformation? Explain 2D – Transformation with suitable examples.

4. Describe the viewing Transformations.

5. Explain 3-D matrix representation.

6. What is projection? Explain the two types of projections.

7. Describe Z-buffer algorithm.

8. Write a short notes on menu-driven command language.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Write the Bresenham's Line-drawing algorithm. To generate line points from (0,0) to (15,15) using Bresenham's line-drawing algorithm.

10. What is tablets? Explain the various types of tablet device.

11. What is clipping? Explain Sutherland line clipping algorithm with an example.

12. Explain the windows and viewing Transformation.

13. Differentiate between two dimensional transformation and three dimensional Transformation. Describe the composite three dimensional transformation.

14. What is coherence? Explain scan line coherence algorithm.

15. Describe painters algorithm with suitable example.

DE–2808

Sub. Code
301

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2015.

Third Semester

DISCRETE MATHEMATICS

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Show that

$$\left((P \vee Q) \wedge \neg\left(\neg P \wedge \left(\neg Q \vee \neg R\right)\right)\right) \vee \left(\neg P \wedge \neg Q\right) \vee \left(\neg P \wedge \neg R\right)$$

is a tautology.

2. Show that $R \to S$ can be derived from the premises $P \to (Q \to S)$, $\neg R \vee P$ and Q.

3. Define the following with an example :

   (a) Proper subset

   (b) Universal set

   (c) Equal sets

   (d) Empty set.

4. If $X = \{1, 2, 3, 4\}$ and

   $R = \{\langle 1,1\rangle, \langle 1,4\rangle, \langle 4,1\rangle, \langle 4,4\rangle, \langle 2,2\rangle, \langle 2,3\rangle, \langle 3,2\rangle, \langle 3,3\rangle\}$. Write the matrix of R and sketch its graph.

5. Show that $\sim\sim A = A$.

6. Define the following with an example :

   (a) Subgroup

   (b) Monoid

   (c) Subsemigroup

   (d) Submonoid.

7. Show that every cyclic group of order n is isomorphic to the group $\langle Z_n, +_n \rangle$.

8. Define the following with an example :

   (a) Graph

   (b) Connected graph

   (c) Directed graph

   (d) Sub graph.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. (a) Obtain the principal disjuctive normal form of

   $P \to ((P \to Q) \wedge \neg(\neg Q \vee \neg P))$.

   (b) Obtain the principal conjuctive normal form of

   $(P \to (Q \wedge R)) \wedge (\neg P \to (\neg Q \wedge \neg R))$. (8 + 7)

10. (a) Prove that

   $(\exists x)(P(x) \wedge Q(x)) \Rightarrow (\exists x)P(x) \wedge (\exists x)Q(x)$.

   (b) Show the following by constructing derivations

   $(x)(P(x)) \to (Q(y) \wedge (R(x)))$,

   $(\exists x)P(x) \Rightarrow Q(y) \wedge (\exists x)(P(x) \wedge R(x))$. (8 + 7)

11. (a) If $S=\{1,2,\cdots,10\}$ and a relation R on S where $R=\{\langle x,y\rangle / x+y=10\}$. What are the properties of the relation R?

(b) If A be a set of factors of a particular positive integer m and if $\leq$ be the relation divides i.e.,

$\leq = \{\langle x, y\rangle / x \in A \wedge y \in A \wedge \ (x \text{ divides } y)\}$.

Draw Hasse diagrams for

(i) $m=2$

(ii) $m=6$

(iii) $m=30$

(iv) $m=210$

(v) $m=12$ and

(vi) $m=45$. (8 + 7)

12. (a) If $f(x)=x+2$, $g(x)=x-2$ and $h(x)=3x$ for $x \in R$, where R is the set of real numbers. Find $g \circ f$; $f \circ g$; $f \circ f$; $g \circ g$; $f \circ h$; $h \circ g$; $h \circ f$; and $f \circ h \circ g$.

(b) Show that

$f(A \cup B)=f(A) \cup f(B)$; $f(A \cap B) \subseteq f(A) \cap f(B)$. (8 + 7)

13. (a) Prove that the kernel of a homomorphism g from a group $\langle G,*\rangle$ to $\langle H,\Delta\rangle$ is a subgroup of $\langle G,*\rangle$.

(b) Show that in a group $\langle G,*\rangle$, if for any $a,b \in G$, $(a*b)^2=a^2*b^2$ then $\langle G,*\rangle$ must be abelian. (8 + 7)

14. (a) If g be a homomorphism from a group $\langle G, * \rangle$ to a group $\langle H, \Delta \rangle$ and if K be the kernel of g and $H' \subseteq H$ be the image set of g in H. Then prove that G/K is isomorphic to H'.

(b) Show that every subgroup of a cyclic group is normal. (8 + 7)

15. (a) In a simple digraph, prove that the length of any elementary path is less than or equal to $n-1$, where n is the number of nodes in the graph.

(b) How many different directed trees are there with three nodes? How many different ordered trees are there with three nodes? (8 + 7)

DE–2809

Sub. Code
302

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2015.

Third Semester

COMPUTER NETWORKS

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. List out the applications of Computer Network.

2. Discuss about Fiber optics cable.

3. Explain various error detection methods.

4. What are the different types of communication mode in HDLC? Explain.

5. Explain Link state routing.

6. Explain the duties of the transport layer.

7. Explain the functions of SNMP.

8. Write a short notes on Electronic mail.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Describe the Network Architecture with neat diagram.

10. Explain the Elementary datalink protocols.

11. Discuss about collision free protocols.

12. Explain the frame format for token ring and token bus.

13. Explain any two adaptive algorithm with suitable example.

14. Discuss about congestion control.

15. Explain the www in detail.

DE–2810

Sub. Code
303

DISTANCE EDUCATION

M.C.A. (S) / M.C.A. (L) DEGREE EXAMINATION, DECEMBER 2015.

Third Semester

SOFTWARE ENGINEERING

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. What are project planning activities? Explain.

2. Write short notes on software requirement specification principles.

3. What are the fundamental principles of system design? Explain.

4. What is data dictionary? How it is useful in structured analysis?

5. Differentiate between cohesion and coupling.

6. What do you mean by interface design? Explain.

7. Explain the benefits and problems with software reuse.

8. Distinguish between verification and validation.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. What is a software process? Explain the fundamental activities common to all software processes.

10. Explain any one of the evolutionary software process model with neat diagram.

11. What is a behavioral model? Draw the data flow diagram for an order processing system and explain.

12. (a) What are SCM standards? Explain.

    (b) What is mile stone? Why it is been used in software development?

13. What is software design? Explain in detail about various design methods.

14. Explain the software testing strategies.

15. Discuss the following

    (a) Basis path testing

    (b) Control structure testing.

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DE–2811

Sub. Code
304

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2015.

Third Semester

VISUAL PROGRAMMING

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Describe the various datatypes available in VB programming.

2. Explain the data control in VB.

3. Differentiate between input box( ) and message box ( ) with suitable example.

4. What is scroll bar? Explain the properties and modules.

5. Describe the MDI form in VB programming.

6. Explain how the static splitter window is created in VC++.

7. Differentiate between COM and DCOM in details.

8. Write down the steps to create the connection with database MS-ACCESS.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. What is an array? Implement a VB program to check an element is present in an array or not. How do you increase the size of an array in VB? Explain.

10. What is common dialog box control? Explain all the methods and properties with examples.

11. List some VC++ resources. Explain the creation and use of a menu resource in detail.

12. (a) Explain the various window common dialog boxes available in VC++.

    (b) Explain how to give connectivity to a data source using VC++ in details.

13. (a) Explain the different document view architecture component of VC++.

    (b) Explain the process of event handling in VC++.

14. What are DLL? Write a sample program to create a user defined DLL.

15. What is object linking and embedding (OLE)? Discuss in detail how it is supported in VC++.

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DE-2812

Sub. Code
401

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2015.

Fourth Semester

ACCOUNTING AND FINANCIAL MANAGEMENT

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. What are the advantages of double entry system?

2. Journalise the following transactions in the books of Tmt. Amutha and prepare the ledger accounts also.

2004			Rs.
Jan.	1,	Tmt. Amutha commenced business with cash	50,000
Jan.	2,	Purchased goods for cash	10,000
Jan.	5,	Purchased goods from Mohan on credit	6,000
Jan.	7,	Paid into bank	5,000
Jan.	10,	Purchased furniture	2,000
Jan.	15,	Sales goods for cash	7,000
Jan.	20,	Sold goods to Suresh on credit	5,000
Jan.	25,	Cash sales	3,500
Jan.	26,	Paid to Mohan on account	3,000
Jan.	31	Paid salaries	2,800

3. From the following Balance sheet, calculate :

   (a) Current ratio

   (b) Liquid ratio

   (c) Debt-Equity ratio

   (d) Proprietary ratio.

Balance Sheet

	Rs.		Rs.
Share capital	5,00,000	Fixed assets	14,00,000
Reserves	3,00,000	Stock	5,00,000
6% debentures	11,00,000	Debtors	2,00,000
Bank overdraft	1,00,000	Cash	1,00,000
Creditors	2,00,000		
	22,00,000		22,00,000

4. What is return on investment? Explain its different aspects?

5. From the given particulars find out the following variances.

   (a) Material price variance

   (b) Material usage variance

   (c) Material cost variance.

   Quantity of material purchased – 3,000 units

   Value of materials purchased – Rs. 9,000

   Standard quantity of material required per tonne of finished products is 25 units.

   Standard rate of material : Rs. 2 per unit

   Opening stock of material : Nil

   Closing stock of material : 500 units

   Finished production during the year : 80 tonnes.

6. What are the managerial uses of variance analysis?

7. From the following figures, prepare raw materials purchase budget.

	Material (units)			
	A	B	C	D
Estimated opening stock	16,000	6,000	24,000	2,000
Estimated closing stock	20,000	8,000	28,000	4,000
Estimated consumption	1,20,000	44,000	1,32,000	36,000
Standard price per unit (Rs)	0.25	0.05	0.15	0.10

8. From the following estimates, calculate the average amount of working capital required :

	P.a. (Rs.)
(a) Average amount locked – up in stocks	
Stock of finished goods and W.I. P	10,000
Stock of stores, materials, etc	8,000
(b) Average credit given :	
Local sales : 2 weeks credit	1,04,000
Sales out side the state : 6 weeks credit	3,12,000
(c) Time available for payment	
For purchases : 4 weeks	78,000
For wages 2 weeks	2,60,000
Add 10% to allow for contingencies.	

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. From the following Trial balance and other particulars, prepare trading and profit and loss account for the year ended 31.12.2010 and a balance sheet as on that date.

Debit balance	Rs.	Credit balance	Rs.
Interest	1,500	Ram capital	40,000
Rent	500	Bills payable	4,000
Taxes and insurance	200	Sundry creditors	20,000
Plant and machinery	20,000	Bank over draft	20,000
Sundry debtors	20,000	Loan on mortgage of premises	30,000
Wages	30,000	Sales	96,400
Stock (1.1.2010)	5,000		
Salaries	7,000		
Bills receivable	3,000		
Loose tools	21,000		
Cash in hand	1,000		
Business premises	40,000		
Stocks of books and stationary	18,500		
Office expenses	700		
Purchases	42,000		
	2,10,400		2,10,400

Adjustments :

(a) Closing stock was Rs. 8,000

(b) Wages outstanding Rs. 600

(c) Salaries unpaid Rs. 200

(d) Rent due Rs. 150

(e) Insurance prepaid Rs. 50

(f) Bad debts written off Rs. 200.

10. The following are the Balance sheets of S.V.P Co. Ltd as on 31.12.95 and 31.12.1996.

Balance Sheets

Liabilities	1995 Rs.	1996 Rs.	Assets	1995 Rs.	1996 Rs.
Share capital	1,00,000	1,25,000	Building	1,00,000	95,000
General reserve	25,000	30,000	Machinery	75,000	85,500
Profit and loss a/c	15,250	15,300	Stock	50,000	37,000
Bank loan	35,000	–	Debtors	40,000	31,100
Creditors	75,000	67,600	Cash	250	300
Provision for tax	15,000	17,500	Bank	–	4,000
			Investment	–	2,500
	2,65,250	2,55,400		2,65,250	2,55,400

Additional information :

(a) Dividend of Rs. 11,000 was paid

(b) Machinery was purchased for Rs. 15,000

(c) Income tax paid during the year Rs. 16,500

Prepare cash flow statement.

11. The sales and profit for 1996 and 1997 are as follows :

		Sales Rs.	Profit Rs.
1996	–	1,50,000	20,000
1997	–	1,70,000	25,000

Find out :

(a) P/V ratio

(b) BEP

(c) Sales for a profit of Rs. 40,000

(d) Profit for sales of Rs. 2,50,000 and

(e) Margin of safety at a profit of Rs. 50,000.

12. Draw up a flexible budget for a production at 75% and 100% capacity on the basis of the following data for a 50% activity.

		Per unit Rs.
Materials	–	100
Labour	–	50
Variable expenses	(direct)	10
Administrative expenses	(50% fixed)	40,000
Selling and distribution expenses	(60% fixed)	50,000
Present production	(50% activity)	1,000 units

13. What are the factors that determinants of working capital requirements?

14. Project X initially costs Rs. 25,000. It generates the following cash flows :

Year	Cash inflows Rs.	Present value of Re. 1 at 10%
1	9,000	.909
2	8,000	.826
3	7,000	.751
4	6,000	.683
5	5,000	.621

Taking the cut-off rate as 10% suggest whether the project should be accepted or not.

15. What is flexible budget? State the importance of flexible budget?

DE–2813

Sub. Code
402

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2015.

Fourth Semester

INTERNET PROGRAMMING

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Describe the security and encryption issues in the web design.

2. Define thread and multithreading. Explain the various states of a thread.

3. Explain the string, math and date objects in JavaScript.

4. Briefly explain the decision making statements available in VBScript.

5. Explain any eight text formatting tags available in HTML.

6. Write a short note on SGML.

7. How do you create a plug – ins? Explain.

8. How do you add graphics to web pages? Explain.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Compare and contrast java with CGI – Perl.

10. Briefly discuss the various looping statements available in JavaScript.

11. Explain any ten buit – in functions available in VB – Script.

12. Describe the form and control tags in HTML.

13. Define web browser. Explain the various web browsers available in the web.

14. Write a detailed note on ActiveX controls.

15. Write a note on :

    (a) Images and hyperlinks

    (b) Dynamic graphics

DE– 2814

Sub. Code
403

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2015.

Fourth Semester

OBJECT ORIENTED ANALYSIS AND DESIGN

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Explain the various events contributed to the evolution of object oriented concepts.

2. Briefly explain the Role of classes and objects in Analysis and Design.

3. List out the various sources used in classical approach. Explain.

4. Explain the process the thread Modelling in detail.

5. What are the criteria to be considered in functional semantics?

6. Explain how objects are mapped with relational system?

7. Write notes on GRASP patterns.

8. Explain the relationships that are possible among the classes in UML representation with your own example.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Discuss the importance of Incremental and Interactive nature of classification.

10. Explain the different types of programming paradigms in the object model.

11. Explain the various methods to measure the quality of an abstraction.

12. Describe the steps to create a domain model used for representing conceptual classes.

13. Write a note on

    (a) Adapter singleton.

    (b) Observer pattern.

14. Discuss the conceptual model of UML.

15. Illustrate with an example about the package diagram and the state diagram.

DE–2815

Sub. Code
404

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, DECEMBER 2015.

Fourth Semester

COMPILER DESIGN

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Briefly explain NFA with an example.

2. Write a note on book keeping.

3. List out the advantages of LR parser over other parsers.

4. Briefly explain ambiguous grammar with an example.

5. Write a note on type checking.

6. Give a brief note on lexical phase errors.

7. Briefly explain the principle sources of optimization.

8. Write a note on DAG for the basic block.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Give the structure of a complier. Explain its phases.

10. Explain the use of table management and error handling.

11. Explain how to construct canonical LR parsing table.

12. Give the model of predictive parser and explain with an example.

13. Translate the expression $(a + b) * (x + y) + (x + y + a)$ into

    (a) Quandruples (5)

    (b) Triples (5)

    (c) Indirect triples. (5)

14. Explain the different data structures used for storing the contents of a symbol table.

15. Discuss the various peephole optimizations that can be done on a source program.

DE–2816

Sub. Code
501

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2015.

Fifth Semester

DISTRIBUTED COMPUTING

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. What do you mean by internetworking? Explain.

2. Differentiate synchronous and asynchronous communication.

3. Write a note on digital signatures.

4. Give a brief description about logical time and logical clocks.

5. Mention the locking rules for nested transaction.

6. Briefly discuss about resource sharing.

7. Write short notes on events and notifications.

8. Briefly explain flat file service interface.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Mention the challenges in distributed system. Explain.

10. Explain :

    (a) UDP datagram communication. (8)

    (b) Group communication. (7)

11. Discuss about different cryptographic algorithms.

12. Explain Bully algorithm with an example.

13. Discuss optimistic concurrency control in detail.

14. Explain the advantages and disadvantages of distributed systems.

15. How to synchronize the physical clock? Explain the different methods in it.

DE–2817

Sub. Code
502

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2015.

Fifth Semester

WEB TECHNOLOGY

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Explain the ordered and unordered list tags in HTML.

2. Write a short note on CSS.

3. What is Java Beans? List the benefits of Java Beans.

4. Describe the servlet's life cycle.

5. What is session management? Explain.

6. Explain the architecture of JSP.

7. Describe the JSP implicit objects.

8. How do you deploy Java Beans in a JSP page?

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain the simple API for XML and DOM.

10. Discuss the date and string objects in JavaScript.

11. What is EJB? Explain the session and entity beans.

12. How do you read quesy string and parameters from HttpServletRequest? Explain.

13. List and explain the interfaces, classes and exceptions available within the Javax.Servlet package.

14. Explain the following :

    (a) Sharing data between JSP pages.

    (b) Sharing session and application data.

15. How do you access a database from a JSP page? Explain.

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DE–2818

Sub. Code
503

DISTANCE EDUCATION

M.C.A./ M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2015.

Fifth Semester

.NET FRAMEWORKS

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. List the advantages of .NET framework.

2. What happens when compile and run a program in .NET? Explain with neat diagram.

3. Define Assembly. Explain the various types of Assembly available in .NET framework.

4. Briefly explain the CTS and CLS.

5. Briefly explain the Marshalling process in .NET.

6. List the various .NET file types.

7. List and explain various controls available in windows forms.

8. Write a note on web service description language.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain the architecture of .NET framework.

10. Define system string. Discuss the various methods available within system. string with example.

11. Write a note on :

    (a) Threads

    (b) Server object types

    (c) Remoting.

12. Explain the ADO .NET object model.

13. Discuss the following :

    (a) Exposing web services

    (b) Consuming web services.

14. How do you display, edit and sort data using DataGrid control? Explain.

15. Describe the basic web controls and their class hierarchy.

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DE–2819

Sub. Code
504

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2015.

Fifth Semester

DATA MINING AND WARE HOUSING

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Define data mining. Differentiate data mining from traditional data processing.

2. Write the difference between data base and knowledge base.

3. List the various classifications of data mining systems.

4. Describe the approach to decision tree pruning.

5. Justify the need for tools to manage a data warehouse.

6. Explain the data warehouse and its architecture.

7. What is meta data? List its advantages.

8. Discuss briefly about testing the data warehouse.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. "Data mining is a multi – disciplinary field" – Justify.

10. Give the architecture of a typical data mining system. Discuss the major issues in data mining system.

11. Discuss about the classification methods using decision tree with an example.

12. Explain the steps involved in knowledge discovery system and data pre-processing techniques in detail.

13. What are the various OLAP operations that can be performed on multi – dimensional data model?

14. List and explain the steps involved in designing star flake schema.

15. Write short notes on :

    (a) Backup strategies

    (b) Testing strategies

    (c) Disaster recovery.

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DE–2820

Sub. Code
601

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2015.

Sixth Semester

MOBILE COMMUNICATIONS

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Enumerate the different kinds of applications of mobile communications.

2. Explain the significance of GSM.

3. What is meant by MAC? Why is it necessary in the wireless communication?

4. Discuss the pros and cons of Wireless networks.

5. What are the different kinds of technologies that can be used to set up WLAN? Explain any one of them.

6. Discuss the basic architecture of cellular IP.

7. State the differences between TCP and UDP.

8. What are the different components of WAE? Explain two of them.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. What are the different kinds of MAC? Explain any three of them in detail.

10. What is meant by Handover? How it could be classified? Explain.

11. What are the basic differences between wireless WANs and wireless LANs? State the common features between them.

12. (a) What is meant by collision? In what situations can collisions occur in all the networks?

    (b) Distinguish between collisions on PHY and MAC layer. How do the wireless networks try to solve the collisions or minimise the probability of collisions?

13. List out the various requirements for a mobile IP and justify them. Does mobile IP fulfill them all? Explain.

14. (a) What are general problems of mobile IP regarding security and support of quality of service? Discuss.

    (b) Briefly explain the reverse tunneling.

15. Write short notes :

    (a) WAP Architecture

    (b) Wireless Telephony.

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DE–2821

Sub. Code
602

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2015.

Sixth Semester

MIDDLEWARE TECHNOLOGY

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Define middleware. List out the types of client server architecture.

2. Write a note on DOTWET architecture.

3. List and explain the XML syntax rules in detail.

4. Compare COM and CORBA.

5. Define IIS. What is the use of ASP.NET runtime?

6. Discuss about the ASP.NET validation controls.

7. Write a note on SOAP.

8. Discuss the ASP.NET framework.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain the different kinds of client – server models with example.

10. (a) Discuss the JDBC derives.

(b) Write a note on RMI.

11. How do you use 'page class' in EJB? Explain with example.

12. (a) Explain the use of CMP and BMP in EJB.

(b) Write a note on session bean.

13. (a) Explain the structure of ASP.NET page.

(b) Discuss the page execution model in ASP.NET.

14. Explain all Rich control in ASP.NET with example.

15. (a) Discuss Gridview control in ADO.NET.

(b) Explain connected and disconnected environment in ADO.NET.

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DE–2822

Sub. Code
603

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2015.

Sixth Semester

AGENT BASED INTELLIGENT SYSTEM

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Define AI. Explain the history of artificial intelligence.

2. Write a short notes on functionalities of agent problem.

3. Explain the characteristics of knowledge based agent.

4. Briefly explain the frames with suitable example.

5. List the functionalities of continuous planning agent.

6. Explain the semantic nets.

7. Describe the probabilistic reasoning.

8. Define learning. Explain its.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain in detail about the breadth first search and depth – first search.

10. Explain the following algorithms :

    (a) A* algorithm

    (b) Mini – Max algorithm.

11. Discuss in detail about the unification algorithm.

12. Give a detailed note on :

    (a) Conditional planning

    (b) Multiagent planning.

13. Explain the following terms :

    (a) Baye's theorem

    (b) Dempster – Shafer theory.

14. Compare and contrast monotonic reasoning and non – monotonic reasoning.

15. Explain about the various statistical learning methods.

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DE–2823

Sub. Code
604

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2015.

Sixth Semester

OPEN SOURCE ARCHITECTURE

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Discuss about the history of Linux.

2. List the advantages of open source software.

3. Explain about the software installation in Linux administration.

4. How do you compile and run C, C++ and Java program in Linux operating system. Give examples.

5. Write a note on Firewalls.

6. Explain sed script with example.

7. Explain any two debugging tools in GNU with examples.

8. Write a brief note on the looping structures in python programming.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Give a detailed note on file and directory permission in Linux.

10. Explain computation in shell programming with example.

11. Describe about the file system organisation and explain how to perform a system backup and restoring process in Linux environment?

12. Write short notes on :

    (a) File compression

    (b) Web browsers

    (c) E – mail services.

13. Explain about various GNU compiler tools.

14. What is GNU libraries? Explain static and dynamic linked libraries in GNU.

15. Explain about X – Windows server architecture in detail.

DE–9061

Sub. Code
105

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, DECEMBER 2015.

First Semester

DATA STRUCTURES AND USING C LAB

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

Examiner has to select ONE question by the lot system and give it to the student.

1. (a) Write a C program to implement the ADT operations in First In First Out (FIFO) method using an array. (50)

   (b) Write a C program to implement the matrix multiplication. (40)

   Record. (10)

---------- Cut here ----------

2. (a) Write a C program to search an element using binary search methods. (50)

   (b) Write a C program to find the factorial of a given number using recursion. (40)

   Record. (10)

3. (a) A file of employees contains data (eno, name and salary) for each employee of a company. Write a program to do the following :

   (i) Create a file

   (ii) Insertion in a file

   (iii) Deletion from a file

   (iv) Modification in a file. (55)

   (b) Write a C program to reverse a given number. (35)

   Record. (10)

---------- Cut here ----------

4. (a) Write a C program to implement the insertion and deletion operations on Stack using linked list. (55)

   (b) Write a C program to sort a set of elements using Selection sort. (35)

   Record. (10)

---------- Cut here ----------

5. (a) Write a C program to implement the insertion and deletion operations using a doubly linked list. (50)

   (b) Write a C program for matrix multiplication. (40)

   Record. (10)

6. (a) Write a C program to implement the insertion and deletion operations in Last In First-Out Method (LIFO) using an array. (50)

(b) Develop a C program to sort N numbers in ascending order using an array. (40)

Record. (10)

---------- Cut here ----------

7. (a) Develop a menu driven program to search an element using binary search methods. (55)

(b) Write a C program to show the sum of N elements of array and show the average. (35)

Record. (10)

---------- Cut here ----------

8. (a) Write a C program to implement the push and pop operations using linked list. (55)

(b) Develop a C program to find whether the given number is prime or not. (35)

Record. (10)

DE–9062

Sub. Code
106

DISTANCE EDUCATION

M.C.A. (S) DEGREE EXAMINATION, DECEMBER 2015.

First Semester

RELATIONAL DATABASE MANAGEMENT SYSTEMS LAB

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

Examiner has to select ONE question by the lot system and give it to the student.

1. (a) Consider the following database consisting of the following tables :

Employee (ssn, first name, last name, gender, designation, doj, address)

Employee – salary (ssn, basic pay, DA, TA, pay)

Department (did, dname, mgrssn)

Employee – department (ssn, deptid)

Employee – dependency (ssn, depname, depgender, dep relationship)

Queries :

(i) Retrieve the names of employees who have no dependents.

(ii) Retrieve all the information about employees working in 'Research' department including the department information.

(iii) Display the department having employee count > 5.

(b) Write a PL/SQL cursor to do payroll processing with employee – salary table.

2. (a) Consider the following database consisting of the following tables.

Employee (employee name, street, city)

Works (employee name, company name, salary, doj)

Company (company name, city)

Manager (Employee name, manager name)

Queries :

(i) Find the names, streets and cities of residence of all employees who work for 'first bank corporation' and earn more than 10,000.

(ii) Find the employees in the database who live in the same cities as the companies for which they work.

(iii) Find those companies whose employees earn a higher salary on average than the average salary at 'First Bank Corporation'.

(b) Write a PL/SQL function to return the name of the employee in a given company name.

3. (a) Consider the following database consisting of the following tables :

Employee (employee name, street, city)

Works (employee name, company name, salary, doj)

Company (company name, city)

Manager (employee name, manager name)

Queries :

(i) Find all employees in database who live in the city 'Chennai' and under the manager 'John'.

(ii) Find all employees who earn more than average salary of all employees of that company.

(iii) Find the number of employees in each company.

(b) Write a PL/SQL procedure to find employee name whose company name as 'Small bank Corporation'.

4. (a) Consider the following database consisting of the following tables :

Department (dept id, dept name)

Student (roll no, name, gender, mark 1, mark 2, mark 3, total, average, dept id)

Staff (staff id, name, designation, qualification, dept id)

Tutor (roll no, staff id)

Queries :

(i) Display the student details who come under the tutor ship of the given staff name 'X'.

(ii) Display the student details who got greater than overall average marks of their department.

(iii) How many students are there in CSE department?

(b) Write a trigger to display the message "Invalid marks" when marks are < 0 and > 100 and to calculate and update total and average when a record is inserted.

5. (a) Consider the following database consisting of the following tables :

Branch (bname, bcity, assets)

Account (ano, starting date, balance)

Customer (cusid, name, address)

Deposit (ano, cusidm bname)

Transaction (ano, amount, mode, date of trans)

Queries :

(i) Find the average account balance at each branch and display only if it is greater than 10000.

(ii) Display the branch details located in a city starting with the letter 's'.

(iii) Find the number of depositors in each branch.

(b) Write a view with the account table for the balance greater than 25000 and delete the record with balance 28000 from the view.

---------- Cut here ----------

6. (a) Consider the following database consisting of the following tables :

Party (pid, pname, leader)

Constituency (cid, cname)

Contestant (ctid, ctname, ctaddr)

Election (ctid, number of votes, pname, cname)

Queries :

(i) Display the contestant details if they secured greater than 10,000 votes.

(ii) Find the number of contestants, constituency wise.

(iii) Display the winner details in each constituency.

(b) Write a PL/SQL function to return the number of votes secured for a given contestant name.

7. (a) Consider the following database consisting of the following tables :

Inventory (item, level, cost)

Minlevel (item, level)

Reorder (item, quantify)

Purchase (item, quantity, cost customer name, date of purchase)

Queries :

(i) Display the number of customers for the shop on a particular day.

(ii) Write a query to display the item purchased by a given customer name.

(iii) Display the overall income for the shop on a given date.

(b) Write a view to display item and their reorder level and minimum level of that item.

Update the recorder level of item × to 10 the view.

8. (a) Consider the following database consisting of the following tables :

Hostel (hno, hname, type [boys/girls]

Menu (hno, day, breakfast, lunch, dinner)

Warden (wname, qual, hno)

Student (sid, sname, gender, year, hno)

Queries :

(i) Display the total number of students in the particular hostel.

(ii) Change the breakfast of the hostel 5 on Thursday to 'Noodles'.

(iii) Display the wardens for each hostel with the qualification 'B.Com.'

(b) Write a PL/SQL function to display the students residing in a given hostel no.

DE–9063

Sub. Code
205

DISTANCE EDUCATION

M.C.A. (S) DEGREE EXAMINATION, DECEMBER 2015.

Second Semester

C++ AND GRAPHICS LAB

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

Record : 5
Algorithm/Flow chart : 15
Program : 35
Execution : 35
Result : 10
Total : 100

Examiner has to choose ONE question for each candidate by lot system.

Each subdivision carries 50 marks.

1. (a) Write a C++ program to draw a rectangle and fill it with four colours equally.

   (b) Write a C++ program to draw a circle using Bresunham's algorithm.

---------- Cut here ----------

2. (a) Write a C++ program to apply transformation principles for a triangle.

   (b) Write a C++ program to draw a straight line using simple DDA technique.

3. (a) Write a C++ program to draw any 3-D object and perform animation on it.

   (b) Write a C++ program to create a circle and make the circle to move on the straight path.

---------- Cut here ----------

4. (a) Write a C++ program to draw the object square.

   (b) Develop a C++ program to draw object triangle divide the object into two quadrants and fill each quadrant with a colour.

---------- Cut here ----------

5. (a) Write a C++ program to implement curve clipping algorithm using graphic functions.

   (b) Write a procedure and a C++ program to draw non-linear object curve.

---------- Cut here ----------

6. (a) Write the procedure and a C++ program to draw object square and four concentric square.

   (b) Write a C++ program to draw a object.

---------- Cut here ----------

7. (a) Write a C++ program to draw a rectangle and in triangle that co-insides with the base of the triangle.

   (b) Write a C++ program to draw the 3-D cube object and perform animation on it.

8. (a) Develop a C++ program to draw non-linear object over.

   (b) Write a C++ program to draw a ellipse with Bresunhams algorithm.

---------- Cut here ----------

9. (a) Develop a C++ program to draw a curve line using simple DDA technique.

   (b) Write a C++ program to show graphical transformation rotation of a 2-D object.

---------- Cut here ----------

10. (a) Develop a C++ program to draw a square and fill it with eight colours equally.

    (b) Write a C++ program to draw a circle and perform the movement of the circle from one end to other end.

DE–9064

Sub. Code
206

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, DECEMBER 2015.

Second Semester

UNIX AND SHELL PROGRAMMING LAB

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

Each candidate should select ONE question using Lot system.

Each subdivision carries 50 marks.

1. (a) Write a shell program to get the input file name and do the following test.

   (i) File existence

   (ii) File readable or not

   (iii) Executable or not.

   (b) Write a shell program to find the sum of first 100 even numbers.

---------- Cut here ----------

2. (a) Write a shell program to perform the following tasks.

   (i) Display all files in currency directory

   (ii) Display only a readable files.

   (b) Write a shell program to convert all capital letters in a file to small letters.

3. (a) Write a shell program which accepts the names of a file from the standard input and then perform the following test on it.

   (i) Enter the 5 names in a file

   (ii) Sort the name in existing file

   (iii) List unsorted and sorted file

   (iv) Quit.

   (b) Write a shell program to sum up the series.

---------- Cut here ----------

4. (a) Write a menu driven shell program to copy, edit, rename and delete a file.

   (b) Write a shell program to find a factorial of a given numbers.

---------- Cut here ----------

5. (a) Write a shell program to evaluate the following series $\frac{1}{2!}+\frac{2}{4!}+\frac{3}{6!}+\ldots+\frac{5}{10!}$.

   (b) Write a shell program to display the message like "Welcome to Unix" when user login.

---------- Cut here ----------

6. (a) Write a shell program to display the result 'PASS' or 'FAIL' using the information given below.

   Student name, Reg.No., Mark 1, Mark 2, Mark 3, Mark 4 the minimum pass for each subject is 50.

   (b) Display the list of last 20 files present in the current directory. Also store this list in a file profile.

7. (a) Write a shell program to prepare electricity bill in the following format for domestic consumers (make own assumptions).

   (i) Consumer number

   (ii) Consumer name

   (iii) Previous reading

   (iv) Current reading

   (v) Unit consumed

   (vi) Number of units consumed and the amount.

   (b) Write a shell program to prepare pay slip for a employee (make relevant field).

DE–9065

Sub. Code
305

DISTANCE EDUCATION

M.C.A/M.C.A (Lateral) DEGREE EXAMINATION, DECEMBER 2015.

Third Semester

NETWORK LAB

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

Break up of marks

Record Note Book : 5

Algorithm/Flow chart : 15

Program : 35

Debugging and Execution : 35

Result : 10

Total : 100

Examiner has to choose ONE question for each candidates by lot system.

Each subdivision carries 50 marks.

1. (a) Write a Java program to find the sum of n natural numbers.

   (b) Write a Java program to read a file from the remote system.

2. (a) Write a Java program to multiply two numbers using RMI client /server technique.

   (b) Write a Java program to setup a chat server and chat client.

---------- Cut here ----------

3. (a) Write a Java program to find a factorial of given number using RMI client/server technique.

   (b) Write a Java program to set up time server and time client.

---------- Cut here ----------

4. (a) Write a Java program to find a given number is Armstrong or not using RMI client/server technique.

   (b) Write a Java program to display the users and their IP address logged in the network.

---------- Cut here ----------

5. (a) Write a Java program to find the given number is prime or not using RMI client/server technique.

   (b) Write a Java program to ENCRYPT a given word.

---------- Cut here ----------

6. (a) Write a Java program to add two numbers using RMI client/server technique.

   (b) Write a Java program to setup echo server and echo client.

7. (a) Write a Java program to find a factorial of given number using FACTORIAL INTERFACE technique.

   (b) Write a Java program to send message from one system to another system.

---------- Cut here ----------

8. (a) Write a Java program to find a given number is prime or not using RMI client/server technique.

   (b) Write a Java program to ENCRYPT and DECRYPT a given word.

DE–9066

Sub. Code
306

DISTANCE EDUCATION

M.C.A/M.C.A (Lateral) DEGREE EXAMINATION, DECEMBER 2015.

Third Semester

VISUAL PROGRAMMING LAB

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

Examiners has to select ONE question and give it to the student by lot system.

Answer both (a) and (b).

1. (a) Write a VB program for string operations using string functions, input box and command button.

   (b) Write visual C++ win 32 application program using MFC that create a new font.

---------- Cut here ----------

2. (a) Write a VB program using control array that creates a scientific calculator with appropriate command buttons. Include the following capabilities for the calculator: + , –, *, /, %, power , square root , square and log (base 10).

   (b) Create a windows using C++ source file.

---------- Cut here ----------

3. (a) Write a VB program for student mark sheet preparation using data report.

   (b) Write visual C++ win 32 application program using MFC that displays current mouse coordinates in status bar.

4. (a) Write a VB program using Active X control to create a text box that accepts only numeric value. It should include back color, fore color and text.

   (b) Write a visual C++ program to create a database connectivity using MFC application Wizard.

---------- Cut here ----------

5. (a) Write a VB project for commercial bank operations using SB account database, with the following features :

   (i) ADD NEW ACCOUNT

   (ii) DEPOSIT AMOUNT

   (iii) WITHDRAW AMOUNT (with minimum balance condition)

   (b) Write a VC++ program to access the student database through MFC ODBC.

---------- Cut here ----------

6. (a) Write a VB program to create a student database (Roll No, St Name, sex, course, DOB, mark 1, mark 2, mark 3, mark 4, mark 5, total, results) using MS Flex grid control.

   (b) Write a visual C++ program to create document-view-frame-menu using APPwizard.

DE–9067

Sub. Code
405

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2015.

Fourth Semester

INTERNET PROGRAMMING LAB

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

Each subdivision carries 50 marks.

ONE question is to be given by the examiner to each candidate by Lot system.

1. (a) Create a simple calculators using Javascript and HTML.

   (b) Write a Java program to insert few records into student table and to display the details of MCA fourth semester students in ascending order (Name-Wise).

---------- Cut here ----------

2. (a) Create a web page for a newspaper using frames, ordered and unordered list, images, image and text as hyperlinks.

   (b) Write a Java program to insert, delete, modify and view the details of employees working in your college.

3. (a) Design a login page for a company validate the user name and password using VB script.

   (b) Create an application to load an image in applet and rotate the image to 90° and 180°.

---------- Cut here ----------

4. (a) Write a VBScript program for the following :

   (i) Prompt the user for entering a string.

   (ii) Display the given string in Uppercase.

   (iii) Reverse the given string.

   (iv) Check whether the given string in Palindrome or not.

   (v) Find the count of a character 'b' in the given string.

   (b) Write an applet program to draw dots at random locations.

---------- Cut here ----------

5. (a) Write a java program to create a tabbed panels with tool tip text and short cut keys. Each tab should display the subjects of each semester in MCA course.

   (b) Design and publish a web page for your college using HTML, Javascript and VB Script.

---------- Cut here ----------

6. (a) Develop a web page for job recruitment agency using HTML and Javascript.

   (b) Write a java program to view the details of students using Grid controls.

7. (a) Write a javascript to validate the textbox as follows :

   (i) the textbox should not be leaved as empty

   (ii) the value entered in the text box should not exceed 8 characters.

   (iii) numbers are not allowed in the textbox.

   (b) Write a Java program to create a menu driven application for insert, delete, edit and update the students details.

---------- Cut here ----------

8. (a) Write a Javascript to create a color palet and display the text in the color chosen from the color palet.

   (b) Create an applet for the following :

   (i) to display the date and time.

   (ii) to draw circle, line and square.

   (iii) to display an image.

   (iv) to display a dialogue menu.

DE–9068

Sub. Code
406

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2015.

Fourth Semester

COMPILER DESIGN LAB

(2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

Break-up marks :

Record Note Book : 5

Algorithm/Flow chart : 15

Program : 35

Debugging and Execution : 35

Result : 10

Total : 100

Examiner should select and give ONE question to each candidate by lot system.

Each subdivision carries 50 marks.

1. (a) Write a C program to remove comment lines from the given source code.

   (b) Write a C program to convert the infix $(a + b)/(b - c)$ into postfix.

---------- Cut here ----------

2. (a) Write a C program to check whether the following production rule is operator grammar or not.

   $S \rightarrow aBCd$

   $S \rightarrow aBdC.$

   (b) Write a C program to draw transition diagram for the following Regular Expression $(a / b)(a / b)$.

3. (a) Write a C program to check whether a given string is allowed identifier or not.

(b) Write a C program to construct the shift reduce parser for the grammar

$S \rightarrow CC$

$C \rightarrow CC$

$C \rightarrow d.$

---------- Cut here ----------

4. (a) Write a C program to find leading and trailing sets for the grammar

$S \rightarrow a / \wedge / (T)$

$T \rightarrow T, S / S.$

(b) Write a C program to recognize constant in the given source code.

---------- Cut here ----------

5. (a) Write a C program to construct predictive parsing table for the production rule

$S \rightarrow iCt\, ST / a$

$T \rightarrow eS / E$

$C \rightarrow b.$

(b) Write a C program to remove spaces in a given source code.

---------- Cut here ----------

6. (a) Write a C program to check whether the given string is derived from the production rules

$S \rightarrow AbC$

$A \rightarrow aCb$

$C \rightarrow a$

$C \rightarrow Cba.$

(b) Write a C program for DFA to accept the regular expression $(a * b)^*$.

7. (a) Write a C program to produce line numbers in the given source code.

   (b) Write a C program to convert infix expression $a - b * c$ into postfix.

---------- Cut here ----------

8. (a) Write a C program to translate the expression. $(A + B) * (C + D)$ into three address code, quadruples and triples.

   (b) Write a C program to recognize keywords in a given source code.

DE–9069

Sub. Code
505

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2015.

Fifth Semester

.NET LAB

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

Examiner should select and give ONE questions to each candidate by lot system.

Each subdivision carriers 50 marks.

1. (a) Write a VB.NET program for the following :

   (i) To read data from text file

   (ii) To write data to text file

   (iii) To delete a file from your system.

   (b) Develop a web application for advertising various models of mobiles available in your mobile shop using Ad Roatators Control.

---------- Cut here ----------

2. (a) Write a program to insert, edit, delete and view the details of students using disconnected data set.

   (b) Develop a splash screen to show a message to the students of our university regarding their examination dates, vacation, semester fees for every 3 minutes.

3. (a) Write a program to generate a pay slip of employees using crystal report.

   (b) Design a web page for your college using image and wizard controls.

---------- Cut here ----------

4. (a) Write a VB .NET program to display the details of all the students in your class using Grid control.

   (b) Develop a web site for your college that makes use of master pages.

---------- Cut here ----------

5. (a) Write a VB .NET program to create a simple text editor.

   (b) Develop a web application for a book shop using cookies concept.

---------- Cut here ----------

6. (a) Create a customer table. Insert few records into customer table. List the name and address of the customers living in Chennai using Crystal report.

   (b) Write a ASP .NET program to develop a login page that validates the user name and password of each customer in the bank.

---------- Cut here ----------

7. (a) Write a VB .NET program to fetch a student data from the student table and display the fetched student details using Gridview control.

   (b) Develop a web page for your home town that make use of web services and AdRotator control.

DE–9070

Sub. Code
506

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2015.

Fifth Semester

WEB TECHNOLOGY LAB

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

Examiner should select and give ONE question to each candidate by lot system.

Answer both (a) and (b)

1. (a) Create a web page for your district using HTML, Javascript and VB script.

   (b) Write a servelt program to send E-mail message.

---------- Cut here ----------

2. (a) Write a program in Javascript to animate an image.

   (b) Write a ASP .NET program to view the details of customer using Repeater control.

---------- Cut here ----------

3. (a) Write an XML document to display the details of your home town. Write an XSL style sheet and attach it to the XML document. Validate the document using DTD.

   (b) Create a login form using JSP and validate the username and password using javascript at client-side and jsp at server-side.

4. (a) Write a servelt program that demonstrates session tracking using hidden form fields by displaying the shopping cart for a bookshop.

   (b) Create a web page for a mobile shop using HTML, Javascript and JSP.

---------- Cut here ----------

5. (a) Write a ASP .Net program using AdRotator to advertise a mobile shop, super market, photo studio and travels company.

   (b) Write a servlet program to send response to the browser using Generic servlet class.

---------- Cut here ----------

6. (a) Write a Javascript program to display the details of web browsers and operating system in which it runs.

   (b) Write a ASP .Net program to bind the list box with students name and the Dropdown list with students city.

---------- Cut here ----------

7. (a) Create a login form using ASP .Net and validate the customer name and password using compare validator and required field validator.

   (b) Write a JSP program to prepare student performance evaluation document.

---------- Cut here ----------

8. (a) Write a JSP program to insert, delete, modify and view the details of students studying in MCA fourth semester and staying in the hostel.

   (b) Create a dynamic e-card generator using ASP .Net.

DE–9071

Sub. Code
605

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2015.

Sixth Semester

OPEN SOURCE PROGRAMMING LAB

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

Examiner should select and give ONE question to each candidate by Lot system.

Answer both (a) and (b)

1. (a) Write a PHP program to create a student table in MYSQL.

   (b) Create a simple shopping cart application for a mobile shop using PHP.

---------- Cut here ----------

2. (a) Write a PHP program for the following

   (i) To store the user information in a session.

   (ii) To display the information stored in a session.

   (iii) To destroy the session.

   (b) Write a PHP program to insert data into a customer table in MYSQL.

3. (a) Write a PHP program to access data from a student table.

   (b) Write a program to prepare a mark sheet using PHP and HTML.

---------- Cut here ----------

4. (a) Write a PHP program for the following.

   (i) Create a student table in MYSQL.

   (ii) Insert data into student table.

   (b) Write a PHP program for the following.

   (i) Delete a record from the student table where fees-paid column is 'No'.

   (ii) Update a record in the student table where age is greater than 20 and city is 'Karaikudi'.

   (iii) Select all the data from the student table and display it in descending order (name wise).

---------- Cut here ----------

5. (a) Write a shell program to change the extension of a given file.

   (b) Create a simple shopping cart application for TV shop using PHP.

---------- Cut here ----------

6. (a) Write a PHP program to upload a file into server.

   (b) Write a PHP program to create a directory and to read the contents of a directory.

7. (a) Write a PHP program to create an Electricity bill using PHP and MySQL.

(b) Write a PHP program to reverse a given string.

---------- Cut here ----------

8. (a) Write a PHP program to create a web page for online mobile shopping.

(b) Write a PHP program to create a cookie and set the cookies expiration date to 10 days.

DE–9072

Sub. Code
606

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2015.

Sixth Semester

MULTIMEDIA TOOLS LAB

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

Break-up marks

Record : 5
Algorithm/Flow chart : 15
Program : 35
Execution : 35
Result : 10
Total : 100

Examiner has to choose any ONE question for each candidate by lot system.

1. (a) Develop an Animation to rising the sun in the sea.

   (b) Animate your own text to create the Magnifier effect.

---------- Cut here ----------

2. (a) Develop an Animation in FLASH to tween the shapes.

   (b) Animate your name such that only one character is highlighted at one time.

3. (a) Develop an Animation in FLASH to create the morphing effect.

   (b) Implement a program in Maya to create the Class Room using Rendering.

---------- Cut here ----------

4. (a) Develop a Home page using Dream viewer for your favourite Textile Show Room.

   (b) Implement a program in FLASH for getting Rain Fall effect.

---------- Cut here ----------

5. (a) Create a Movie such that the text "FLASH" moves along with the Mouse Cursor.

   (b) Develop a Home Page using Dream Viewer for the Mobile Show Room.

---------- Cut here ----------

6. (a) Animate your own text to create the vertical flipping effect.

   (b) Implement a program to create a Maze using Maya.

7. (a) Implement a program in FLASH to represent the Animation in Time line.

   (b) Develop the program in Maya to create a Chess Board.

---------- Cut here ----------

8. (a) Animate your own Text to create the Rotating effect in Clock-wise Direction and print the Text in Reverse Format.

   (b) Develop an Animation for Butterfly with Three flowers in the garden.